Safety

BluePrint Automation is not and does not represent itself as a safety expert. It is the responsibility of the owner, employer, or equipment user to take all necessary steps to guarantee the safety of all personnel in the workplace. According to **ISO 14121** *Safety of Machinery***,** the owner or user is advised to consult the latest standards to ensure complete compliance to safety Requirements for packaging machinery, packaging-related converting machinery, and robotic systems. As the owner, employer, or user of a packaging machine, it is your responsibility to arrange for training of operator(s) and maintenance of machinery to recognize and respond to known hazards associated with your machinery and to be aware of the recommended operating procedures for your particular application and machinery installation. **BluePrint** Automation therefore, recommends that all personnel who intend to operate, program, repair, or otherwise use the machinery be trained in an approved **BluePrint** Automation training course and become familiar with the proper operation of the system. Persons responsible for programming the system must be familiar with the design, implementation, and debugging of application programs.

Safety

General Safety Precautions

Please read and follow all of the safety precautions listed below before operating the machine loading system.

- Failure to use, maintain, or service the equipment in accordance with this Operation and Technical Reference Manual and the associated vendor-supplied documentation may impair the safety protection provided by the system.
- A shock hazard exists within the equipment control enclosure(s). There are some exposed 400 V~ and 230 V~ connections within the enclosures. A qualified electrician must secure access to the electrical control enclosure(s) before attempting to operate on, troubleshoot, or repair problems with the system.
- Use one of the emergency stop (E-Stop) pushbuttons in case of emergency when injury to personnel or damage to equipment is imminent.
- Observe all safety precautions at all times to reduce the chance of serious injury or equipment damage
- Wear safety glasses at all times around operating equipment. Check all pneumatic (if applicable), electrical, and mechanical connections periodically for tightness and signs of wear. Replace or tighten, as required, to maintain the safety and integrity of the system.
- Do not operate the packaging machinery system with any of the safety enclosure panels removed. Panels that require a tool to remove are not protected by safety switches.
- Keep your hands clear of the rotating parts and conveyor rollers. There is a risk of getting your hands or fingers pinched in the mechanisms.

The system includes safety mechanisms to protect the operator from inadvertently placing hands or equipment into the moving parts during operation. If the associated sensors or switches are triggered, an emergency stop will result.

⚠WARNING

Use extreme care around the electrical components, and any moving parts of the machine because of the risk of personal injury.

⚠WARNING

Do not disconnect, defeat, or remove any of the safety devices installed on the machine. Under normal operating conditions, these devices protect the operator and technician from moving or electrical parts. The safety devices include the emergency stop pushbuttons, lockable disconnect switch, safety switches, safety relays, fixed guards, and openable guards.

Signal Word Panels

The convention followed in this manual for signal word panels comports with international standard **ISO 3864** *Graphical symbols -- Safety colours and safety signs*. The signal word panels are deployed as follows:

This signal word panel is used to communicate information about practices not related to personal injury.

This signal word panel indicates a hazardous situation which, if not avoided, ***could*** result in **minor** or **moderate injury**.

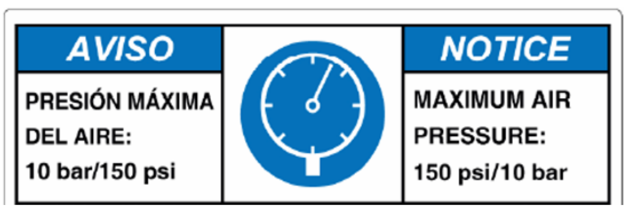
This signal word panel indicates a hazardous situation which, if not avoided, ***could*** result in **death** or **serious injury**.

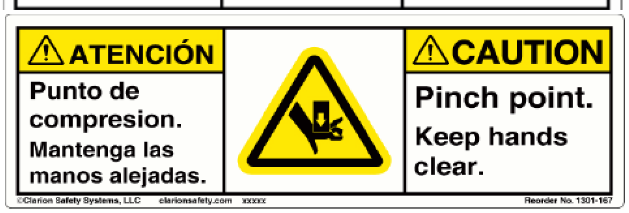
This signal word panel indicates a hazardous situation which, if not avoided, ***will*** result in **death** or **serious injury**.

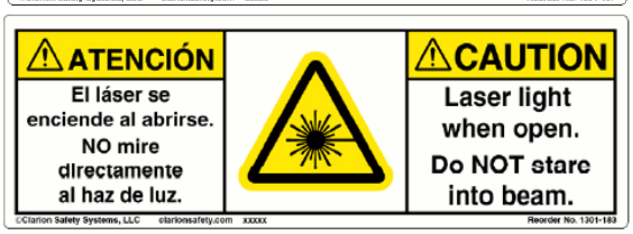
Safety Signs

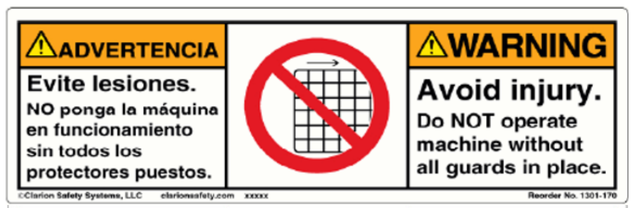
Safety signs are placed on this machine as a constant reminder that the operator must be sharp and alert when operating this machine. Never remove, paint, or deface a safety sign. Should a safety sign be damaged or defaced, contact **BluePrint** Automation for a replacement.

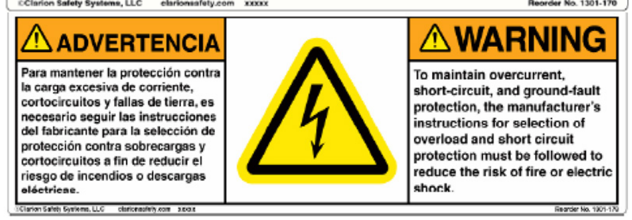
The following safety signs are used on **BluePrint** Automation machines. Review the signs that are on your specific equipment and be familiar with their locations and messages.

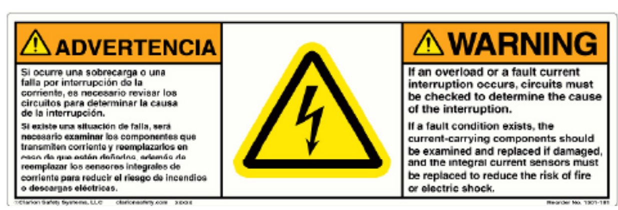


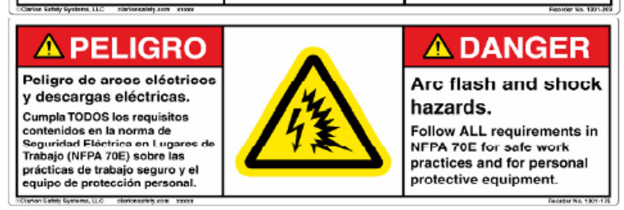












Mechanical Safety

Mechanical

Allow enough area around the machine for operating and cleaning personnel access. Once this equipment is set, permanently fasten the support pads securing the equipment to the floor. Visually check the interior and exterior of this equipment to ensure that all shipping and transport restraints have been removed. Electrical wiring and pneumatic tubing are enclosed inside the frame tubing. Never drill into or weld the frame.

Mechanical Safety

Because of its rotating and fast moving parts, our machines are protected on all sides, usually by transparent doors and panels. Positionable guards are fitted with safety switches which stop the machine when the guard is opened. The product/material infeed and outfeed openings understandably cannot be enclosed and herein lies the danger. Do not insert any limbs in the product/material infeed and outfeed openings while power is applied to the machine.

Mechanical Safety

NOTICE

***BluePrint* Automation does not recommend the use of synthetic oils. Tests have shown that pneumatic component O-rings tend to expand with the use of synthetic oils.**

Operator Guards

When operating this machine all guards must be closed and in their proper locations.

Before operating this machine do the following:

- Check for loose guards.
- Check for missing guards
- Check for broken guards.
- Tighten or replace guards as necessary.

Do not alter or modify any guard.

Physical Contact

When the machine is powered (i.e. not in **Lockout** condition) remember the following:

- Do not lean on the machine
- Do not reach into the machine
- Do not climb on the machine
- Do not crawl under the machine

Minor Maintenance

Place the main power disconnect switch in the “OFF” position and in **Lockout** condition before cleaning, adjusting, or performing maintenance. Before returning the main power disconnect switch in the “ON” position, make certain all guards have been returned to their proper location and are closed.

Electrical Safety

Electrical Safety

Safety Switches

Safety switches are used to render the machine inoperable should an safety panel or guard be opened or removed. Safety switches provide increased protection for operators. For your protection do not override, defeat, or bypass safety switches.

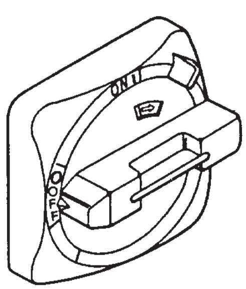
Electrical Panels

Keep electrical power panel enclosures closed at all times. Allow only authorized personnel access into electrical panel enclosures.

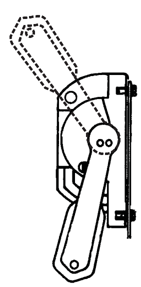
Electrical Supply

Interconnecting plugs and cables are used to install this equipment. Keep all cables off the floor. The main electrical cabinet has a disconnect handle. Placing this handle in the “OFF” position de-energizes the equipment. However, the input side of the disconnect switch is still energized. Turn off and **Lockout** the main power source to the cabinet before servicing.

Typical Main Power Switches



Rotary Type



Lever Type

NOTICE

Electrical Supply: Because our clients have different requirements, gland access holes for installation of the main electrical power are not provided. Be certain wire gauge and insulation meet all applicable code requirements. The power source must have a breaker in a lockable enclosure.

WARNING

Never bypass, strap, or otherwise deactivate a safety device, such as a limit switch, for any operational convenience. Deactivating a safety device is known to have resulted in serious injury and death.

Emergency Stop Pushbutton

All machines are equipped with emergency stop pushbutton(s). If an emergency occurs, the machine or zone can be shut down by simply pressing any one of the emergency stop pushbuttons. When an emergency stop pushbutton is pressed, all motors and belt conveyors lose driving power. In addition air pressure on all pneumatic parts is removed and a service stop signal is transmitted to upstream equipment.

Emergency Stop Pushbutton

Staying Safe During Inspection

Mechanical Installation Inspection

Allow enough area around machine for operating and cleaning personnel. Once this equipment is set, permanently fasten the support pads securing the equipment to the floor. Visually check the interior and exterior of this equipment to ensure that all shipping restraints have been removed. Electrical wiring and pneumatic tubing are enclosed inside the frame tubing. Never drill into or weld the frame.

When inspecting the machine, be sure to:

- Turn off power at the controller.
- **Lockout** and **Tagout** the power source at the Main Power Switch according to the policies of your plant.
- If machine motion is not needed for inspecting the electrical circuits, press the **Emergency Stop Pushbutton** on the operator panel.
- Never wear watches, rings, neckties, scarves, or loose clothing that could get caught in moving machinery.
- If power is needed to check the machine motion or electrical circuits, be prepared to press the **Emergency Stop Pushbutton**, in an emergency.

⚠WARNING

Never bypass, strap, or otherwise defeat a safety device such as a limit switch, for any operational convenience. Defeating safety devices is known to have resulted in serious injury or death.

Staying Safe During Maintenance

Maintenance Safety

⚠WARNING

Do not try to remove any mechanical component from the machine before thoroughly reading and understanding the procedures in the appropriate manual. Doing so can result in serious personal injury and component destruction.

Know the work area of the machine and ***ALL*** guard hoods and/or door opening devices.

- Know the location and status of all switches, sensors, and/or control signals that might cause the machine elements, conveyor, and opening devices to move.
- Make sure that the work area near the machine is clean and free of water, oil, and debris. Report unsafe conditions to your supervisor.
- Become familiar with the complete task the machine will perform BEFORE operating the machinery.
- Never enter the work envelope during automatic operation unless a safe area has been designated.
- Never wear watches, rings, neckties, scarves, or loose clothing that could get caught in moving machinery.
- Stay out of areas where you might get trapped between moving conveyors, or an opening device and another object.
- Be aware of signals and/or operations that could result in the triggering of guns or bells.
- Be aware of all safety precautions when dispensing of paint is required.
- Follow the procedures described in this manual.

⚠WARNING

Never bypass, strap, or otherwise deactivate a safety device, such as a limit switch, for any operational convenience. Deactivating a safety device is known to have resulted in serious injury and death.

Safety During Maintenance

- When a maintenance technician is repairing or adjusting a machine, the work area is under the control of that technician. All personnel not participating in the maintenance must stay out of the area.
- For some maintenance procedures, station a second person at the control panel within reach of the **Emergency Stop Pushbutton**. This person must be knowledgable of the machine and associated potential hazards.
- Be sure all covers and inspection plates are in good repair and in place.
- Never use machine power to aid in removing any component from the machine.
- During machine operations, be aware of the moving elements. Excess vibration, unusual sounds, and so forth, can alert you to potential problems.
- Whenever possible, turn off the main electrical disconnect before undertaking cleaning operations.

Conveyor Safety

Conveyor Safety

- When equipment is interfaced between **BluePrint** Automation equipment and existing equipment, care needs be to paid to the guarding of the abutting equipment.
- If conveyors are installed above exits, passageways, aisles or any other place where people walk, a clearance of at least 6 ft, 8 in shall be provided. If it is necessary to install them with less than the minimum requirement then an alternate exit shall be provided. Also a warning of low headroom must be posted.
- Only trained employees shall be allowed to operate conveyors. Training must include operation for normal and emergency situations.
- Keep the area around the conveyors clear of debris and obstructions.

When working near any conveyor:

- Wear hard hat and safety shoes.
- Tie back (and tuck in) long hair.
- Know the location of the emergency “shut-off” devices and how to use them.
- Do not wear loose clothing or jewellery.
- Do not climb on the conveyors.
- Position yourself so that you are not hit by objects moving down the conveyor.
- Ensure that you can see the conveyor system when you are at the operating controls.
- Ensure that guards are in place for all moving parts of the drive system and in all zones where hazards such as in-running nip, drawing-in, trapping and crushing, friction burns or abrasion are present.
- Locate emergency stop devices near the operator and along the length of the conveyor at approximately 30 metres (100 feet) apart (or closer).
- Ground belts on belt conveyors to prevent static buildup.

Gravity conveyors include those that have rollers, wheels or chutes where objects move by gravity or momentum only.

- Guard pinch points on rollers and wheels and between the conveyor and receiving table.
- Provide adequate guardrails along sides to prevent all objects from falling off.
- Provide retarders (friction areas) if heavy objects are conveyed.
- Ensure there are warning devices near the receiving areas if you cannot see the packages moving on the conveyor.
- Ensure draft checks (fire doors) are installed where conveyors pass through fire walls or floors.

⚠WARNING

Never ride on, step on, or rest on a conveyor system.

⚠WARNING

Do not climb, sit, stand, walk, ride, or touch the conveyor at any time

It is imperative that workers never climb, sit, stand, walk, ride, or even touch the conveyor line at any time. It's common sense, but people tend to get mischievous about it and there are injuries and equipment damage every year due to not following this rule.

- Operate equipment only with all approved covers and guards in place
- Do not load a stopped conveyor or overload a running conveyor
- Ensure that all personnel are clear of equipment before starting
- Allow only authorized personnel to operate or maintain material handling equipment

⚠WARNING

Do not perform maintenance on the conveyor until electrical, pneumatic, hydraulic, and gravity energy sources have been blocked or placed into Lockout condition

⚠WARNING

Keep clothing, body parts, and hair away from the conveyor

Only workers who have been trained should be permitted to operate and perform maintenance on conveyors. This is for two reasons:

1. Safety of the technician: Conveyors can be dangerous to those who do not thoroughly understand the equipment and how to safely work on it.

2. Only trained personnel can really maintain a conveyor to perform at peak efficiency. This isn't necessarily a safety concern, although it can be if subpar conveyor performance causes workers to try to look at it on their own or bypass guards.

- Do not modify or misuse conveyor controls
- Remove trash, paperwork, and other debris only when power is in Lockout/Tagout condition. Ensure that ALL controls and pull cords are visible and accessible
- Know the location and function of all stop and start controls
- Report all unsafe conditions. Jams should be cleared ONLY BY authorized, trained personnel

Energy Lockout

Energy Lockout

BluePrint Automation machines are suitably disposed for placing all energy sources in **Lockout** condition using approved energy blocking devices. Refer to your company's internal procedure for the specific program to be followed for locking out energy sources during maintenance and service activities.

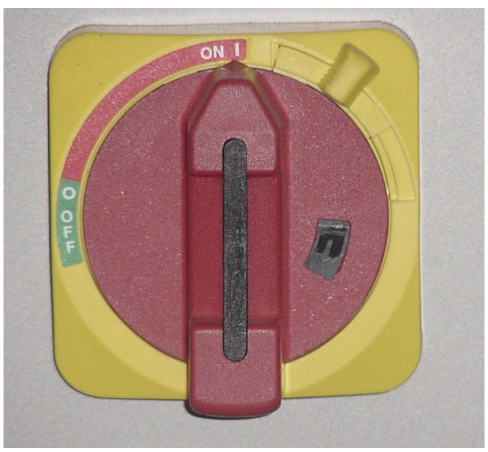
Machine Elements with Lockout Provision

Main Power

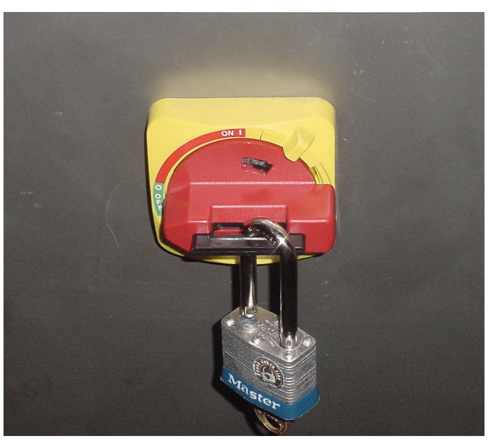
BluePrint Automation uses two different types of Main Power Shut Off Switches. One switch is rotary actuated (A) and the other is lever actuated (B). ***Refer to the section that matches your machine.***

1. To place all electrical power into Lockout condition, move the Power Disconnect to the **OFF** position and align the holes for the lockout device installation. Install an authorized lockout device in the holes so the switch is mechanically blocked from being moved to the **ON** position.
2. Tag the electrical disconnect switch using authorized internal procedures and tags.

A - Rotary Actuated



Front View



Side View

B - Lever Actuated

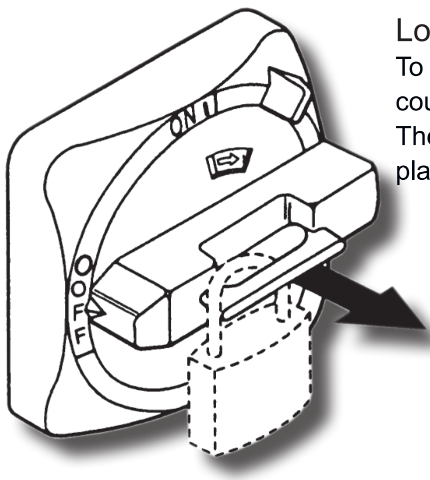

Front View


Side View

NOTICE

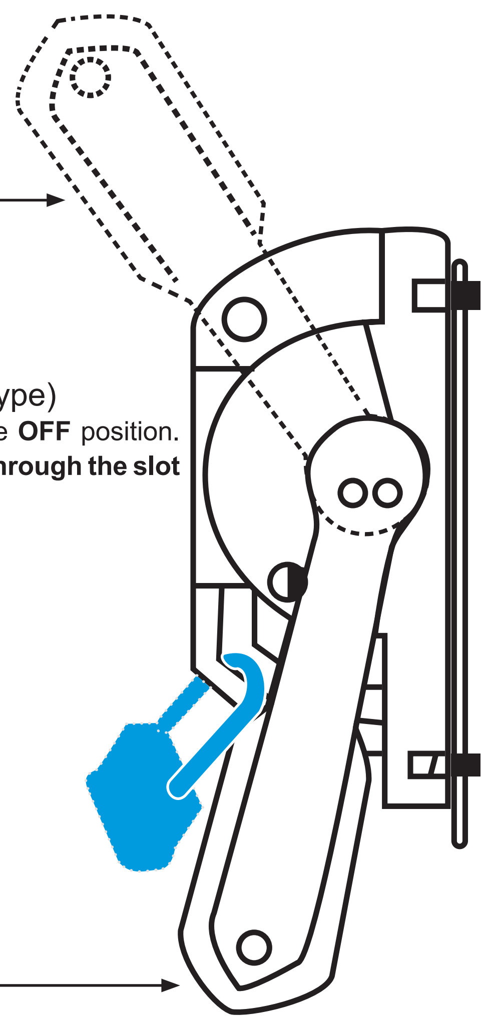
It is the end-user's responsibility to implement a Lockout/Tagout policy in compliance with all relevant safety regulations and legislation in effect in the end-user's geographic location. Lockout/Tagout refers to an energy-control program to ensure that employees isolate machines from their energy sources and render them inoperative before any employee services or maintains them.

When the main power switch is turned off, all components and accessories associated with the system will lose power, (unless special wiring was specified by your company at the time of purchase, in this case the cables will have yellow jackets).



Lockable Main Power Switch (rotary type)

To place the Main Power in **Lockou**t condition, turn the red switch handle counterclockwise to the OFF position.
The Main Power is now OFF. Pull out the black center tab of the handle, and place a lock through the center opening.



Lockable Main Power Switch (lever type)

Move the handle from the on position to the **OFF** position.
The Main Power is now OFF. Place a lock through the slot

General Guidelines for Installing and Removing Lockout/Tagout Devices

Installing Lockout Devices

The following general guidelines, while not a substitute for a well-developed internal **Lockout/Tagou**t policy, should be observed if maintenance or servicing is required around high-voltage components or circuits or on pneumatically actuated equipment:

1. Place electrical power under **Lockout** condition using approved **Lockout** devices. Refer to your company's internal policies for specific **Lockout/Tagout** procedures to be followed for placing equipment under **Lockout** condition during maintenance and service activities.
2. Tag the **Lockout** devices using authorized **Lockout/Tagout** procedures and tags.

Removing Lockout Devices

1. Comply with internal procedures for the removal of **Lockout** devices.
2. Resolve the situation that required the energy sources to be placed under Lockout condition.
3. Make sure the machine and work area are clear of all tools and equipment.
4. Remove **Lockout** device(s) from the pneumatic shutoff valve(s).
5. Rotate the pneumatic shutoff valve handle to the **ON** position.
6. Remove the lockout device from the Electrical Disconnect switche(s).
7. Rotate the Electrical Disconnect switch to the **ON** position.

⚠ DANGER

If it should ever become necessary for personnel to access a powered electrical cabinet, it is the end-user's responsibility to implement a Arc-Flash safety policy in compliance with all relevant safety regulations and legislation in effect in the end-user's geographic location.

Laser Safety

This system incorporates a laser product, the safety information for which is detailed below.

Class 1 Lasers

An IEC 60825-1 Class 1 laser is safe under all conditions of normal use. This means the maximum permissible exposure (MPE) cannot be exceeded when viewing a laser with the naked eye or with the aid of typical magnifying optics (e.g. telescope or microscope). To verify compliance, the standard specifies the aperture and distance corresponding to the naked eye, a typical telescope viewing a collimated beam, and a typical microscope viewing a divergent beam. It is important to realize that certain lasers classified as Class 1 may still pose a hazard when viewed with a telescope or microscope of sufficiently large aperture. For example, a high-power laser with a very large collimated beam or very highly divergent beam may be classified as Class 1 if the power that passes through the apertures defined in the standard is less than the accessible exposure limits (AEL) for Class 1; however, an unsafe power level may be collected by a magnifying optic with larger aperture.

Laser Specifications

Divergence angle: 1 mrad; wavelength: 655 nm; frequency: 7.1 kHz; pulse duration: 0.2 μs; limit value pulse: ≤ 140 mW (IEC 60825-1).